

SynAgent, CADD and Lawrencium

Updated project report: model pipeline, ChEMBL evaluation, correction experiments, and fine-tuning infrastructure

Prepared from the complete working history and the current local artifacts

Report date: August 26, 2026

Project owner: Aaditya Sanil

Repositories reviewed: lukasmki/SynAgent and lukasmki/caddtune

This report replaces the older August 3 document. That earlier report stated that no fine-tuning had occurred; this is no longer true. A 50-step QLoRA smoke run on Lawrencium completed successfully and produced a LoRA adapter.

30.25%

400-path SynAgent/GGUF strict validity

57.0%

500-path validity after order relaxation

14.0%

failing routes repaired to strict validity, n=50

Change-control note

No repository changes were pushed, merged, renamed, or published while creating this PDF. This is a local review artifact for approval.

1. Executive summary

The project now has three proven layers: a working multi-model agent pipeline, a reproducible chemistry benchmark and corrector experiment, and a functioning Lawrencium training environment. The key scientific result is that the original SynLlama validator is strongly sensitive to reactant ordering, so the reported 30.65% baseline is not purely a chemistry score.



What is complete

- SynAgent combines molecule generation, retrosynthesis, validation, and route correction in one pydantic-ai pipeline.
- DeepSeek and Mistral orchestrator support was exercised; DeepSeek was more reliable in the observed trials.
- Tool calls were proven from structured ToolCallPart and ToolReturnPart records, not inferred from final prose.
- The 10,000-path SynLlama baseline was reproduced exactly at 3,065 valid paths (30.65%).
- A 50-route correction study measured a 14.0% strict repair rate and 96.0% tool-chain invocation rate.
- A one-GPU, 50-step QLoRA smoke fine-tune of Llama-3.1-8B completed successfully on an A40.

What remains

- Complete the queued 4xA40 timing run and report 1M, 2M, and full-dataset epoch estimates.
- Run a larger, sampling-matched chemistry benchmark after clarifying the baseline's 'frozen' sampling setting.
- Improve correction chemistry and route normalization before claiming a production-level validity lift.
- Remove the stale overlapping Hugging Face parquet shard or always stage the three source configs separately.
- Rotate the temporary hosted-model API keys used during testing.

2. What changed across the project

Workstream	Changes completed	Verified outcome
SynAgent integration	Combined current model-call capability with corrector architecture; restored deterministic and optional disagreeable prompts; added provider selection.	Agent can generate a molecule, plan a route, validate it, and attempt repairs.
Model serving	Downloaded quantized SmileyLlama and SynLlama GGUF checkpoints; served via llama.cpp; corrected Alpaca prompt format and route token limit.	Local chemistry-model calls produced real outputs through SynAgent.
ChEMBL benchmark	Built resumable harness, copied validator verbatim, added single-instance guard and figures.	Baseline reproduced exactly; 400-path local-model run reached 30.25% strict validity.
Corrector evaluation	Added structural tool-call logging, retries, n=50 harness, screenshots, and revalidation.	48/50 tool chains fired; 7/50 failing routes became strictly valid.
CADDtune	Added synthesis tasks, preference schema, Alpaca-to-chat conversion, and corrected building-block derivation.	Training-set infrastructure passed 28 tests; no CADDtune push in this reporting pass.
Lawrencium	Built probe/setup/staging/pretokenization/smoke/timing scripts; fixed environment and scheduler issues.	50-step QLoRA smoke run completed; 4xA40 timing job queued at last verified state.

Repository state

The active SynAgent development branch is **synagent-full-pipeline**. The requested corrector work is represented there. No direct merge or branch rename is part of this report, and nothing was pushed while generating it.

3. SynAgent full pipeline

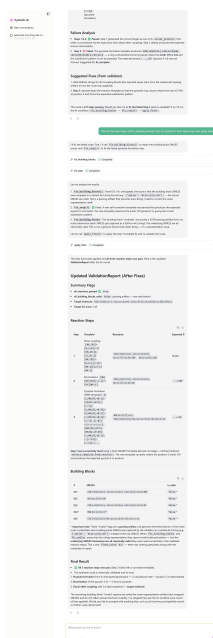
The production path is a tool-using conversation. The orchestrator chooses capabilities; the chemistry models remain specialized tools. The conversation history also acts as the corrector's state store.

Stage	Tool	Role	Evidence
1	generate_molecules	SmileyLlama proposes molecules from property constraints.	Real local GGUF model call.
2	retrosynthesis	SynLlama produces a structured synthesis route.	Real local GGUF model call; max output raised to 1024 tokens.
3	validate_route	RDKit/template checks every block and reaction step.	Structured ValidationReport returned.
4	fix_building_blocks	Attempts replacements for invalid starting materials.	ToolCallPart and ToolReturnPart captured.
5	fix_step	Repairs each failed reaction step.	Failure mode and new template returned.
6	apply_fixes	Assembles the repaired route and revalidates it.	Hard example ended with all three steps passing.

Important implementation fixes

- Chemspace is now loaded only when CHEMSPACE_API_KEY is present; previously both subagents could block all agent construction.
- Seven dataclass capabilities now explicitly annotate id, description, and defer_loading, preventing inherited None/False defaults.
- SmileyLlama prompts use the Alpaca format expected by the 1B checkpoints instead of a Llama-3.1 chat wrapper.
- SynLlama route generation allows 1024 tokens instead of 256, avoiding truncated JSON.
- DeepSeek was added as an OpenAI-compatible pydantic-ai provider; hosted keys were kept in process environment only.

Hard full-pipeline evidence



4. ChEMBL benchmark and validator findings

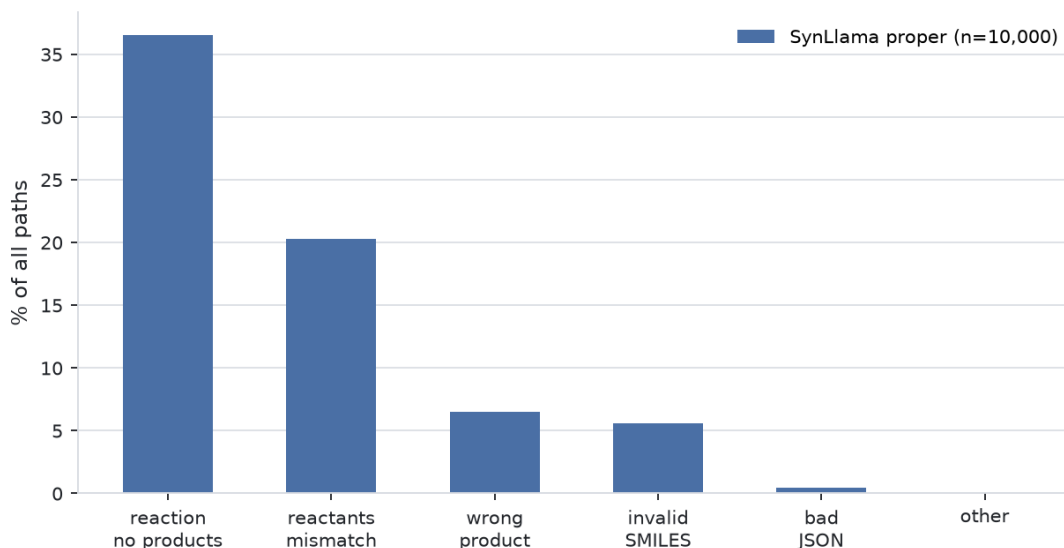
Lukas's data contains 1,000 ChEMBL targets with 10 SynLlama routes per target. The validator was copied verbatim and reproduced the released split exactly: 3,065 valid and 6,935 failed out of 10,000.

3,065 strictly valid baseline routes	6,935 baseline failures	30.65% released baseline validity
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Baseline failure profile

Failure bucket	Count	Percent
Reaction produced nothing	3,654	36.54%
Reactants did not match template	2,027	20.27%
Wrong product	648	6.48%
Invalid SMILES	561	5.61%
Bad JSON / other	45	0.45%

Failure mode breakdown



Local quantized-model pilot

The SynAgent + SynLlama-1B-GGUF run reached 121 valid paths out of 400 (30.25%). This is close to the 30.65% released baseline, but it is not an apples-to-apples model comparison: the checkpoint is a quantized community reproduction and the sampling settings may differ.

5. Headline finding: the benchmark scores reactant order

RDKit RunReactants assigns reactants to template slots positionally. When the same reactants are permuted into a compatible order, many routes previously called invalid pass without any chemical change.

28.2%

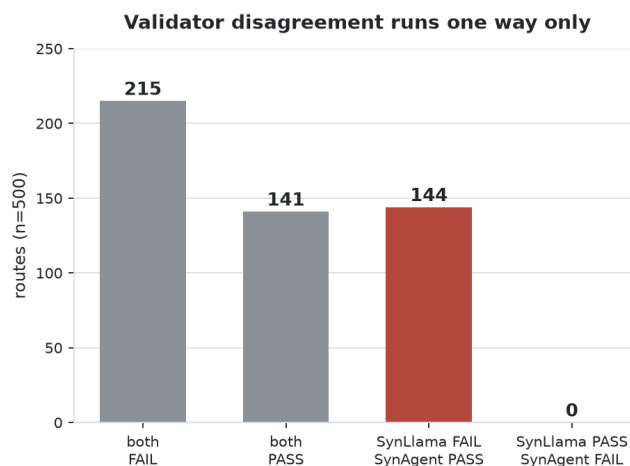
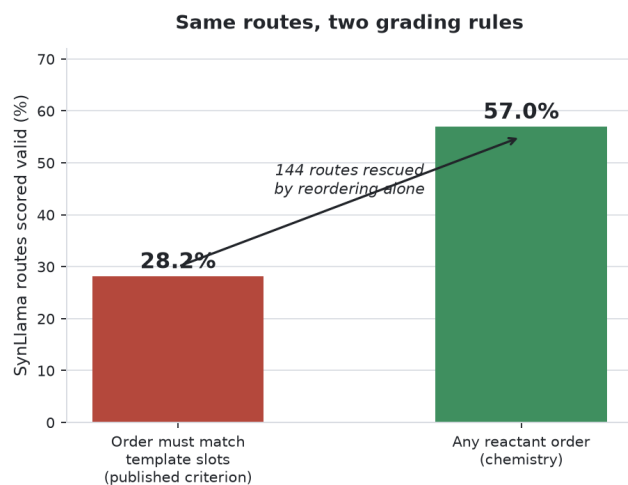
strict order, 141/500

57.0%

best reactant permutation, 285/500

144

routes rescued only by reordering

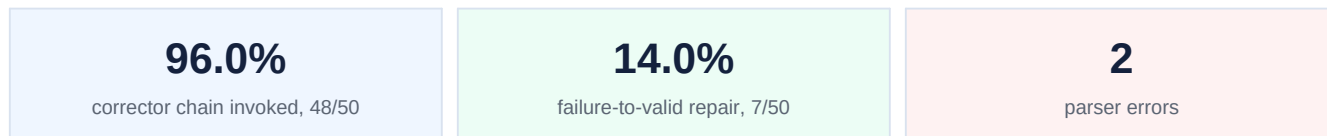


Interpretation

- The published 30.65% score underestimates chemical/template compatibility when reactant order is arbitrary.
- A fair validator should either canonicalize reactants to template slots or test permutations within a safe cap.
- The 57.0% relaxed result is a diagnostic upper estimate on this 500-route sample, not a new official benchmark score.
- Before reporting model improvements, Lukas should clarify what the baseline sampling label 'frozen' means.

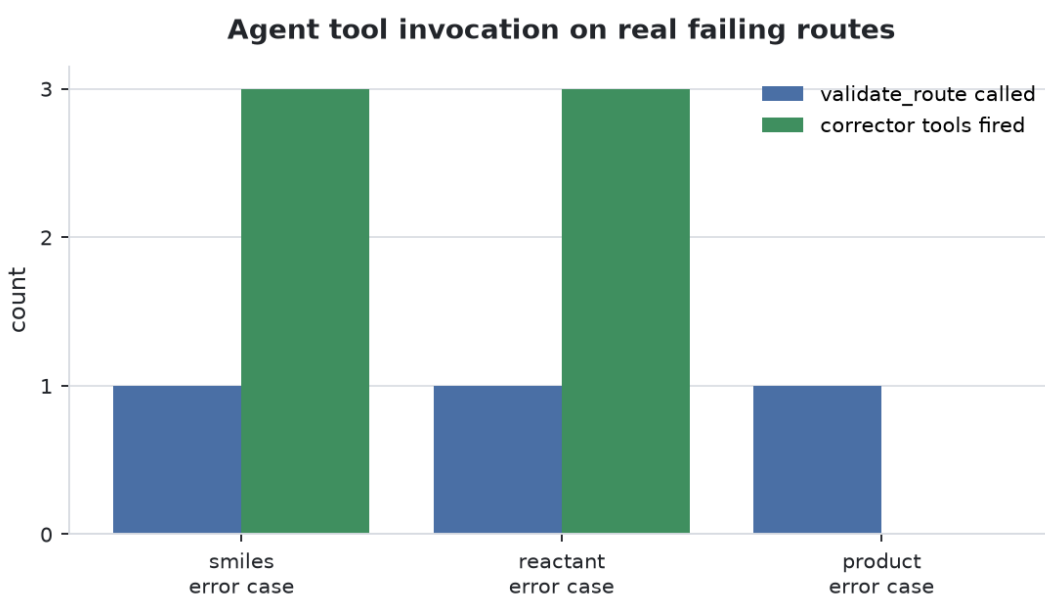
6. Corrector experiment: n=50 failing routes

Fifty routes known to fail the strict baseline validator were sent through the real agent loop. The agent had to call the corrector tools; the repaired output was then rescored with the same validator.



What the number means

Seven of fifty initially failing routes became strictly valid. If that measured rate held across all 6,935 released failures, the overall strict-validity rate would rise from 30.65% to approximately **40.4%**. This is an extrapolation from a small sample, not a completed 10,000-route result; the approximate 95% interval for the repair rate is about 6% to 27%.



Lessons from the run

- Tool invocation was non-deterministic. Explicit repair retries raised invocation from about 75% to 96%, but did not materially raise chemistry success.
- DeepSeek completed 3/3 focused agent cases versus Mistral's 2/3 and avoided the observed Mistral rate-limit failure.
- The corrector is conversation-coupled: it reads validation and prior fixes from message history, so batch evaluation must preserve one full conversation per route.
- A route-representation defect was exposed: some building blocks reach validation as nested dictionary strings rather than plain SMILES.

7. CADDtune and dataset-building work

CADDtune already supported one-source generation into either chat or Alpaca/instruction form. The missing pieces were synthesis-specific training tasks, preference data, and safe multi-step route handling.

Component	Added or corrected	Why it matters
SynthesisGeneration	Target molecule to full ordered route	Creates SynLlama-compatible SFT examples.
SynthesisBuildingBlockGeneration	Target to purchasable leaves	Separates true starting materials from intermediates.
SynthesisReasoning	Retrosynthetic think trace plus answer	Supports reasoning-style chemistry SFT.
SynthesisStepEditing	Template-supervised reaction edits	Implements the missing editing task family.
Preference schema	prompt / chosen / rejected	Enables TRL-style DPO once real pairs exist.
expand_example	Alpaca to conversational chat	Allows format A/B tests on identical content.
Building-block derivation	Only leaf reactants; ordered deduplication	Prevents intermediates being mislabeled as purchasable.

Dataset facts

- The unified cadd-instruct corpus already exists; combining three source datasets is not remaining work.
- The real current three-source dataset contains 5,856,395 rows.
- The Hub 'all' config incorrectly totals 9,712,790 because it globs a stale 3,856,395-row two-source parquet together with the current nine shards.
- Using the broken 'all' config double-weights SmileyLlama and LinkLlama while under-weighting SynLlama.
- The safe staging path explicitly loads smileyllama, linkllama, and synllama configs instead of the overlapping 'all' config.

Validation

The CADDtune training-set layer passed 28 tests with lint and formatting checks clean in the recorded development work. No CADDtune code is being pushed with this report.

8. Lawrencium fine-tuning setup and verified run

The environment, data staging, tokenization, SLURM scripts, and smoke training now work on the lr_ninjaone allocation. Failures were resolved through repeated compute-node tests rather than login-node assumptions.

1 x A40
smoke hardware

50 steps
QLoRA smoke completed

15m 53s
training loop duration

168 MB
saved adapter

Successful smoke result

Job 25143690 completed with exit code 0. PyTorch 2.5.1+cu124 detected one NVIDIA A40. The Llama-3.1-8B QLoRA run completed all 50 steps at roughly 17.89 seconds per step and saved **adapter_model.bin** plus checkpoints 25 and 50 under scratch.

Area	Final verified choice
Partition / account	es1 / lr_ninjaone
QoS	condo_ninjaone_es1; es_lowprio is the preemptible fallback
Environment	Python 3.11 venv, torch 2.5.1+cu124, transformers 4.46.3, trl 0.12.1, peft 0.14.0, axolotl 0.6.0
Training data	1M-row JSONL subset, 1.56 GB
Model cache	25 GB in user scratch
Pretokenization	40-process CPU job, 6m20s, 13 GB cache
Timing job	4 x A40, 28 CPUs, 200 steps, FSDP full-shard; queued at last verified state

Last verified scheduler state

Timing job **25144172** was pending with **QOSGrpNodeLimit**, meaning the condo's two-node cap was occupied. It was intentionally left queued rather than moved to low-priority QoS. The old monitor timed out; that does not cancel the SLURM job.

9. Lawrencium problems found and fixed

Problem	Observed symptom	Resolution
Wrong QoS	es_normal rejected	Use <code>condo_ninjaone_es1</code> or <code>es_lowprio</code> .
Lmod not initialized	python unavailable in batch	Source <code>/usr/share/lmod/lmod/init/bash</code> .
Piped module load	Silently retained Python 3.6	Never pipe module; assert Python 3.11 immediately.
Login /tmp full	Heredocs and scp failed	Stream tar archives over SSH; generate scripts locally.
Windows certificate CRLF	libcrypto unsupported	Strip carriage returns and <code>chmod</code> the key.
Transformers mismatch	Missing <code>AutoModelForVision2Seq</code>	Pin clean transformers 4.46.3 install.
Missing <code>pkg_resources</code>	Import failure	Install <code>setuptools</code> below version 81.
DeepSpeed CUDA probe	<code>CUDA_HOME</code> Missing <code>CUDAException</code>	Remove <code>deepspeed</code> ; use <code>FSDP</code> for the timing run.
CPU preprocess validation	<code>bf16</code> rejected without GPU	Derive CPU-safe config with identical dataset-cache hash.
Excess CPU request	4-GPU job rejected	Use 28 CPUs; 32+ exceeded this allocation's schedulable ceiling.

Operational principle

Downloads, staging, and tokenization were moved away from expensive GPU time. Model and dataset artifacts live under user scratch, while compute nodes perform training. Login nodes are used only for submission and inspection.

10. Claims, limits, and next actions

Safe claims today

- SynAgent demonstrably executes the complete six-tool generate-plan-validate-repair pipeline.
- The released SynLlama validator baseline was exactly reproduced.
- Reactant ordering alone rescued 144 of 500 routes in a controlled diagnostic.
- The corrector converted 7 of 50 strict failures to strict passes, with a 96% tool-chain invocation rate.
- An 8B QLoRA smoke fine-tune completed successfully on one Lawrencium A40.

Claims not yet justified

- Do not claim SynAgent beats official SynLlama: checkpoints and sampling settings were not matched.
- Do not report 40.4% as an observed overall score; it is an extrapolation from n=50.
- Do not treat 57.0% as the new official baseline; it is a permutation diagnostic on n=500.
- Do not claim full-dataset fine-tuning is complete; only the smoke run completed.
- Do not claim every syntactically valid repair is chemically preferable without expert or broader metric review.

Priority next actions

Priority	Action	Deliverable
1	Let timing job 25144172 run when the condo frees; inspect or renew SSH certificate only as needed.	Measured sec/step and projected epoch time for 1M, 2M, and 5.856M rows.
2	Normalize route input and reactant ordering; add regression tests.	A validator score that better reflects chemistry than serialization order.
3	Ask Lukas what sampling_params='frozen' means and reproduce it.	Sampling-matched model comparison.
4	Scale repair evaluation to several hundred failures with error stratification.	Stable repair estimate by failure mode.
5	Fix the overlapping Hugging Face shard or document the safe loader.	Unambiguous 5.856M-row corpus.
6	Rotate the temporary Mistral and DeepSeek keys.	Remove credential exposure risk.

11. Deliverables and evidence map

Artifact	Location / purpose
Status deck	SynAgent-Project-Status.pptx - 21-slide project overview
Benchmark report	chembl-bench/REPORT.md - methods, failures, and screenshots
Benchmark harness	run_bench.py, compare_validators.py, test_order.py
Corrector harness	run_repair.py, verify_tools.py, repair-deepseek-n50.csv
Full-pipeline proof	full-pipeline-hard/hard_attempt1_full.png
Lawrencium runbook	lawrencium/README.md and numbered sbatch/config files
Training output	/global/scratch/users/asanil/runs/smoke/adapt_model.bin
Active SynAgent branch	lukasmki/SynAgent: synagent-full-pipeline

Recorded SynAgent commits already present before this PDF

- 722a4a8 - documentation, ChEMBL benchmark results, agent evidence, cluster scripts, and status deck.
- 93a2b9a - DeepSeek orchestrator support and n=50 corrector repair-rate measurement.

Review note

This report was assembled from the chat history plus the local project report, benchmark outputs, figures, screenshots, Lawrencium runbook, training logs summarized in the working history, and repository artifacts. It is intentionally conservative: measured results, estimates, and pending states are labeled separately.

No changes pushed. No branches merged. No branch renamed.